

Auditing System-1 Models on Biosecurity-Relevant Benchmarks: Calibration, Selective Prediction, and Permutation Instability in a Non-Generative Model

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Abstract—Non-generative “System-1” models return structured probabilistic decisions in a single forward pass, without autoregressive decoding, at a small fraction of the inference cost of a generative model. This makes them of interest as inexpensive components in larger pipelines, but their reliability on biosecurity-relevant tasks is not well characterised, and benchmark accuracy alone does not describe it. We audit one commercial System-1 model on 6,020 multiple-choice items drawn from WMDP, a paraphrase-robust WMDP-Bio variant, and six LAB-Bench subtasks, using 19,061 API calls at an estimated cost of 0.17 USD. We measure accuracy, calibration, error detection, selective prediction, and sensitivity to the order in which answer options are presented. Accuracy is strongly task-dependent. Once the vendor’s uncertainty field is correctly interpreted, the model is reasonably well calibrated (pooled ECE 0.034) and its top-1 probability separates correct from incorrect predictions (pooled AUROC 0.820). Under four cyclic rotations of the answer options, 37.4% of WMDP-Cyber items receive different answers, and errors are concentrated among these items. Averaging probabilities across rotations improves WMDP-Cyber accuracy by +3.8 pp, and applying it only to low-confidence items recovers most of that gain at 2.2 calls per item rather than four.

Index Terms—model evaluation, calibration, selective prediction, robustness, biosecurity, AI safety

1. Introduction

Recent work on efficient inference has produced models that return a typed, probabilistic decision from a single forward pass rather than by decoding tokens autoregressively. Such “System-1” models are offered for settings in which many decisions must be made cheaply. The evaluation reported here cost 0.17 USD for 19,061 calls, which makes repeated evaluation of the same item a practical experimental option rather than a hypothetical one.

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Low-cost probabilistic models are potentially relevant to AI safety and biosecurity pipelines, where auxiliary classifiers are already deployed alongside larger models [1], [2]. Before such a role could be considered, the basic empirical properties of these models need to be established: how accurate they are, how well their reported uncertainty tracks correctness, whether that uncertainty supports abstention, and how stable their decisions are under changes that do not alter the meaning of the input. Benchmark accuracy addresses only the first of these.

This paper reports a reliability audit of one commercially available System-1 model, Jev, on two kinds of benchmark. WMDP [3] provides biosecurity-relevant hazardous-knowledge questions, and LAB-Bench [4] provides practical biology research tasks. Neither benchmark asks the model to judge whether a request is dangerous, so neither measures screening or hazard-detection performance. They are used here as subject matter on which reliability, uncertainty, and robustness can be studied.

We ask four questions. When is the model correct? Does its reported uncertainty identify likely errors? Can uncertain items be set aside to raise accuracy on the remainder? And are its decisions stable under a change that does not affect meaning, namely the order in which the answer options are listed?

Contributions. This paper makes three contributions:

- 1) A reliability audit of a commercial non-generative System-1 model across 6,020 biosecurity- and biology-relevant benchmark items, covering accuracy, calibration, error detection, and selective prediction.
- 2) A controlled answer-order study showing substantial item-level instability despite comparatively stable aggregate accuracy, with errors concentrated among unstable items.
- 3) An evaluation of selective permutation averaging, showing that repeated inference applied only to low-confidence items improves accuracy

on the instability-heavy task while avoiding the cost of evaluating every item four times.

We do not claim novelty for confidence-based selection of items for further computation [5]–[8], nor for the observation that multiple-choice evaluation is sensitive to option order [9], [10].

2. Background and Related Work

Efficient decision models and cascades. Spending additional computation only on inputs a cheap model finds difficult is an established technique. CALM [5] exits a decoder early under a calibrated criterion, Big Little Decoder [6] escalates when a small model’s maximum predicted probability falls below a threshold, and Frugal-GPT [7] and RouteLLM [8] allocate queries across models under a cost budget. Our final experiment applies the same idea, using additional answer permutations of the same model rather than a larger model.

Uncertainty and selective prediction. We use standard tools: the maximum class probability as an error-detection baseline [11], expected calibration error and reliability diagrams [12], and risk–coverage analysis from selective classification [13].

Multiple-choice robustness. Reordering answer options changes model rankings on leaderboards [9] and degrades accuracy in model-dependent ways [10]; formatting changes that preserve meaning can also shift accuracy substantially [14]. That work concerns generative models scored by likelihood over option tokens. We test whether a non-generative model emitting a typed distribution in one pass shows comparable sensitivity.

Safety classifiers and biosecurity context. Deployed language-model systems commonly place auxiliary classifiers around a primary model to detect disallowed content: moderation classifiers [15], input–output safeguard models such as Llama Guard [1] and Shield-Gemma [16], and constitutional classifiers trained to resist jailbreaks [2]. Biological risk is a specific concern within this literature [17], and WMDP [3] was introduced to measure hazardous knowledge so that it can be reduced. This work motivates why inexpensive probabilistic models are of interest, but we do not evaluate any of these safeguard tasks. Prior safeguard work asks whether a system can detect or block a dangerous request; we ask what reliability properties a cheap System-1 model exhibits on biosecurity-relevant knowledge benchmarks.

3. Experimental Setup

Model. Jev is a commercial non-generative model queried through a REST endpoint. A request carries a free-text state and a typed question; the response gives a selected category, a probability distribution over categories, and a scalar confidence. The model is proprietary: no logits, token probabilities, or random seed

are exposed, and returned probabilities are quantised to two decimals (101 distinct values observed). Because inference is a single pass rather than autoregressive decoding, per-item cost is low; we did not measure latency and make no claim about it.

Datasets. Table 1 lists the 10 datasets. WMDP-Bio, WMDP-Chem, and WMDP-Cyber are hazardous-knowledge proxies [3], WMDP-Bio-Robust is a paraphrased variant of WMDP-Bio, and six LAB-Bench subtasks cover practical biology research [4]. Option counts range from 2 to 10, so we report accuracy against each item’s own chance level $1/n$ rather than a single 25% line. LAB-Bench items are formed by combining the reference answer with its distractors under a fixed shuffle seed; items with fewer than two distinct options after de-duplication are skipped. Of 6,021 single-pass calls, 1 returned an API error and is excluded.

Metrics. We report accuracy with Wilson intervals; expected calibration error (ECE) over ten equal-width bins; AUROC for separating correct from incorrect predictions using $p_{\max}$, the largest returned probability; the error rate conditional on high confidence, $P(\text{error} \mid p_{\max} \geq 0.9)$; and risk–coverage curves obtained by withholding the least confident items.

Cyclic evaluation. For the two four-option WMDP suites we additionally present each item under all four cyclic rotations of its option list (1,273 and 1,987 items, 13,040 calls), mapping each returned distribution back to the original option indices so that rotations are comparable.

Cost. At 0.042 USD per million input tokens with no output charge, the complete study cost an estimated 0.17 USD. Token counts are estimated from payload length, as the endpoint reports no usage field.

4. Results

4.1. Benchmark performance and uncertainty

Accuracy varies widely across tasks (Table 1). It is highest on WMDP-Bio (85.2%), falls on the paraphrased WMDP-Bio-Robust (80.1%), and is substantially lower on WMDP-Cyber (63.4%). LAB-Bench results are lower and more variable, from 53.7% on ProtocolQA to 36.6% on SuppQA, close to that subtask’s 0.213 chance level. Aggregate accuracy therefore describes the particular task at least as much as the model.

Interpreting the model’s uncertainty requires first establishing what the reported confidence field measures. Across all 6,020 records it matches

$$c = \frac{p_{\max} - 1/n}{1 - 1/n}$$

to within ± 0.015 for 97.9% of records, where n is the number of options. Competing readings fit less well: raw $p_{\max}$ matches 28.0%, the margin between the top two options 31.0%, and normalised entropy 20.7%. The field

TABLE 1. Per-dataset results. Lift is chance-adjusted accuracy, $(\text{acc} - c)/(1 - c)$, using each item’s own option count $c = 1/n$ rather than a single 25% line, since option counts range from 2 to 10. ECE and AUROC are computed on $p_{\max}$ rather than on the vendor confidence field (§4.1). $P(\text{err} \mid p_{\max} \geq 0.9)$ is the error rate given a confident answer, with Wilson intervals. Retained is accuracy on the items kept when the least confident 40% are withheld (§4.2).

Suite	Dataset	N	opts.	accuracy [95% CI]	lift	ECE	AUROC	$P(\text{err} \mid p_{\max} \geq 0.9)$	retained
WMDP	WMDP-Bio	1273	4	0.852 [0.83, 0.87]	0.802	0.036	0.855	0.039 [0.03, 0.05]	0.970
	WMDP-Bio-Robust	811	4	0.801 [0.77, 0.83]	0.735	0.044	0.823	0.063 [0.04, 0.09]	0.936
	WMDP-Chem	408	4	0.730 [0.69, 0.77]	0.641	0.057	0.824	0.056 [0.03, 0.10]	0.902
	WMDP-Cyber	1987	4	0.634 [0.61, 0.66]	0.512	0.033	0.774	0.075 [0.06, 0.10]	0.787
LAB-Bench	LB-CloningScenarios	32	4–8	0.469 [0.31, 0.64]	0.314	0.126	0.671	0.500 [0.09, 0.91]	0.579
	LB-ProtocolQA	108	4–7	0.537 [0.44, 0.63]	0.397	0.124	0.620	0.077 [0.01, 0.33]	0.600
	LB-SeqQA	600	2–5	0.618 [0.58, 0.66]	0.492	0.093	0.861	0.000 [0.00, 0.05]	0.825
	LB-DbQA	520	4	0.398 [0.36, 0.44]	0.197	0.086	0.653	0.091 [0.03, 0.28]	0.465
	LB-LitQA2	199	2–10	0.402 [0.34, 0.47]	0.208	0.168	0.656	0.381 [0.21, 0.59]	0.479
	LB-SuppQA	82	4–8	0.366 [0.27, 0.47]	0.195	0.193	0.709	0.250 [0.07, 0.59]	0.429
pooled		6020	2–10	0.673 [0.661, 0.685]	–	0.034	0.820	–	–

is a rescaling of the top-1 probability onto $[0, 1]$, mapping a uniform distribution to 0 and a one-hot distribution to 1; on a four-option item an even split between two candidates ($p_{\max} = 0.5$) is reported as 0.33. It is not a predicted probability of correctness, and calibration should be assessed against $p_{\max}$ instead.

On that basis, pooled ECE is 0.034 (95% CI [0.027, 0.045]; Brier 0.161), against 0.091 when the rescaled field is used (Fig. 1(a)). Pooled AUROC for separating correct from incorrect predictions is 0.820 (95% CI [0.810, 0.830]), so $p_{\max}$ carries information about correctness. We compared four candidate signals, $p_{\max}$, the top-two margin, negative entropy, and the rescaled field; they differ by at most 0.006 AUROC, so the choice among them is not consequential.

Calibration quality is not uniform. Per-dataset ECE is at most 0.057 on the four WMDP suites but reaches 0.168 on LitQA2 and 0.193 on SuppQA, where item counts are small.

4.2. Confidence and selective prediction

A quantity that is directly interpretable for a system consuming these predictions is the error rate given a confident answer. Under $P(\text{error} \mid p_{\max} \geq 0.9)$ the model is most reliable on WMDP-Bio (3.9%, $n = 848$) and less so on WMDP-Cyber (7.5%, $n = 550$); a Fisher exact test gives $p = 0.00469$, and $q = 0.0141$ after Benjamini–Hochberg correction across all pairwise comparisons. Confident answers are considerably less reliable on the LAB-Bench literature tasks, reaching 38.1% on LitQA2 (95% CI [21, 59]%, $n = 21$), although that subset is small. A confidence threshold calibrated on one dataset should not be assumed to transfer to another.

Withholding low-confidence items raises accuracy on the remainder (Fig. 1(b)). Deferring the least confident 40% of items raises accuracy on the retained set from 0.852 to 0.970 on WMDP-Bio and from 0.634 to 0.789 on WMDP-Cyber. Confidence therefore identifies a subset on which the model performs substantially better, which

is a prerequisite for any scheme that would direct difficult items elsewhere.

4.3. Answer-order instability

We next ask whether decisions are stable under a change that does not alter the content of the question. Each item is presented under all four cyclic rotations of its option list, and each returned distribution is mapped back to the original option indices.

Aggregate accuracy is nearly unaffected: across the four rotations it spans 0.8 pp on WMDP-Cyber and 1.3 pp on WMDP-Bio. Item-level agreement is much lower. Only 62.6% of WMDP-Cyber items receive the same answer under all four rotations, against 89.4% on WMDP-Bio (Fig. 2(a)). Error rates differ sharply between the two groups: on WMDP-Cyber, 62.5% of unstable items are answered incorrectly under the original ordering against 21.3% of stable items. A substantial fraction of errors therefore occurs on items whose predictions are unstable under reordering.

Two observations qualify this. First, an association between instability and error is partly implied by the measurement: consistency and correctness are both defined relative to the first rotation, and there is one way to be consistently correct but several ways to be wrong. A null model with no positional preference, matched to the same accuracy and the same consistency, produces an error ratio of 5.43 on WMDP-Cyber against the 2.93 observed, so the association is not larger than chance alone would produce. The descriptive result stands; a causal reading does not. Second, the first rotation presents options in the same order as the single-pass run, and the two disagree on 4.03% of WMDP-Cyber items (1.34% on WMDP-Bio). Repeated identical queries do not always return the same answer, so part of the measured instability reflects run-to-run variation rather than option order.

We also examined positional preference directly. Because predicted labels are drawn toward the correct

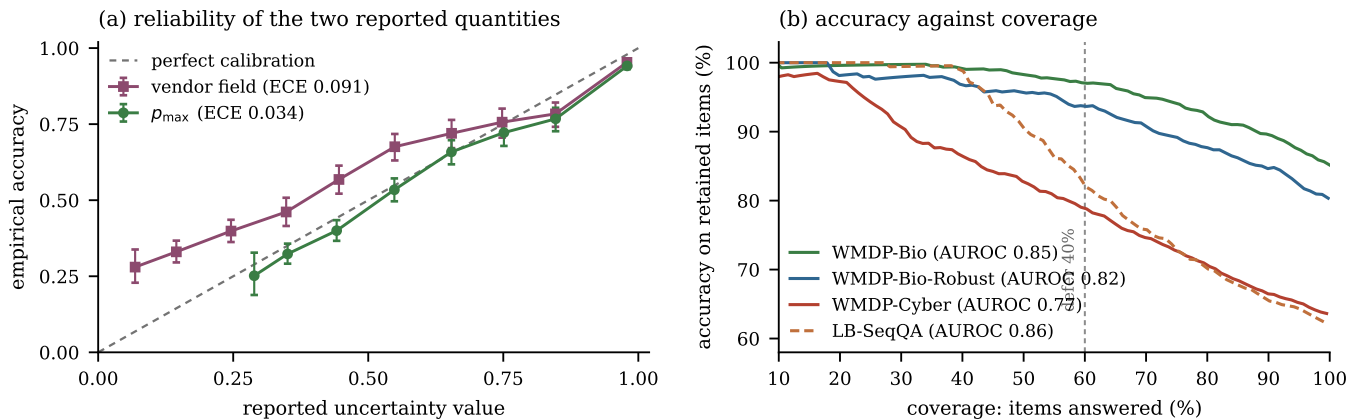


Figure 1. Uncertainty and selective prediction. (a) Pooled reliability curves for the two quantities the API reports. The confidence field is an affine rescaling of $p_{\max}$, so calibration measured against it differs from calibration measured against the probability itself. Bins holding fewer than 25 items are omitted, because $p_{\max} \geq 1/n$ leaves the lowest bins nearly empty; error bars are Wilson intervals. (b) Accuracy on retained items as a function of coverage, withholding the least confident items first.

labels by accuracy, a χ^2 test against the correct-label distribution loses power as accuracy rises. We therefore compare against a simulated null of the same accuracy with no positional preference. The first option is over-selected on 3 of 8 testable datasets, clearly on WMDP-Cyber ($p < 10^{-4}$; 34.8% of predictions against 26.6% of correct answers) but not on WMDP-Bio ($p = 0.11$, power 1.00). On WMDP-Cyber the effect is confined to uncertain items: at $p_{\max} < 0.5$, 42.5% of predictions fall on the first option against 21.8% of correct answers ($p < 10^{-4}$), whereas at $p_{\max} \geq 0.9$ it is absent (27.5% against 27.8%, $p = 0.84$, power 1.00).

4.4. Selective permutation averaging

If a single evaluation returns one of several possible answers, averaging over presentations should help where instability is high. Averaging the four inverse-permuted probability distributions raises WMDP-Cyber accuracy from 0.633 to 0.671 (+3.8 pp, 95% CI [+2.4, +5.3]); majority voting over the four answers is weaker (0.652). On WMDP-Bio, where 89.4% of items are already stable, the gain is +0.5 pp with an interval spanning zero ([−0.2, +1.4]). Averaging helps on the dataset that exhibits substantial instability and not on the one that does not.

Evaluating four rotations costs four times as much as one. Since $p_{\max}$ predicts correctness, it can be used to select which items receive the additional evaluations. Presenting a fraction c of items under all four rotations, chosen least-confident first, costs $1+3c$ calls per item. On WMDP-Cyber, selecting the least confident 20% reaches 0.659 and the least confident 40% reaches 0.670 at 2.2 calls per item, which is 97% of the gain available from averaging every item at 45% lower cost (Fig. 2(b)). Random selection at the same budget reaches 0.648, averaged over 400 draws, and no draw matched the confidence-based ordering. On WMDP-Bio the advan-

tage over random selection is +0.3 pp and 6.8% of random draws do at least as well, so we do not claim a benefit there. The distance to the oracle curve indicates the headroom remaining for a better selection rule.

5. Discussion and Limitations

These results bear on how System-1 models should be evaluated. Aggregate accuracy was nearly invariant across the four rotations while more than a third of WMDP-Cyber items received different answers, so similar accuracy figures can conceal substantial item-level instability. Reported uncertainty is informative once its definition is established: $p_{\max}$ separates correct from incorrect predictions with pooled AUROC 0.820 and supports abstention, but calibration quality and the error rate at high confidence both vary by dataset, so a threshold chosen on one task should not be assumed to hold on another. Stability under re-presentation is a second signal, unavailable from a single evaluation, and repeated inference is worth its cost where instability is high but not where it is low. A direct follow-up is a determinism study: our estimate of run-to-run variation is an incidental by-product of the rotation experiment rather than a designed measurement.

For systems in which users can influence how a decision is presented, representation sensitivity could become a manipulation surface, and disagreement across re-presentations could serve as a signal that a decision is unreliable. Our benchmark does not test such a deployment, and we make no claim about it.

These properties are relevant if System-1 models are later considered as components of biosecurity classifiers or language-model safety pipelines. Evaluating such a use would require a dedicated benign-versus-hazardous classification benchmark, which this study does not provide.

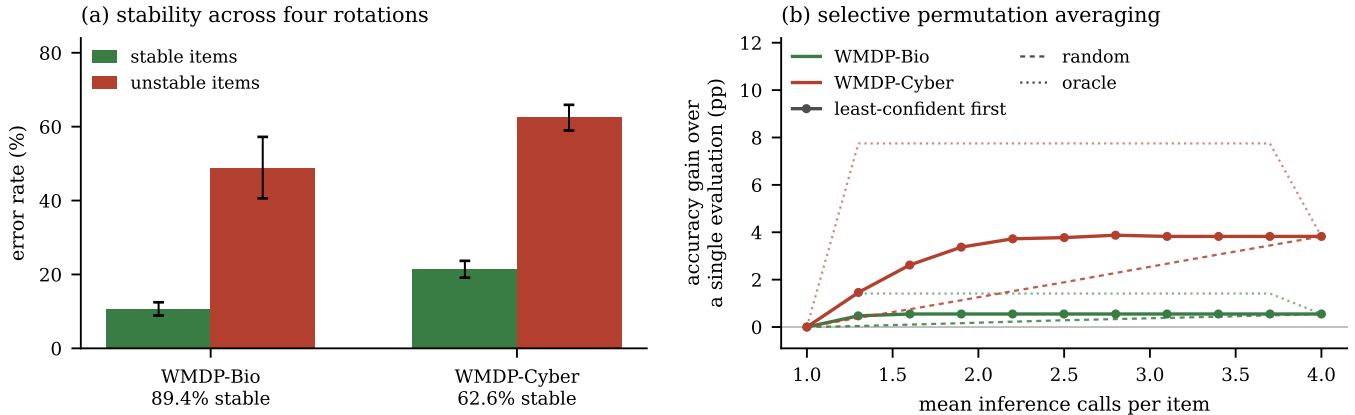


Figure 2. Answer-order instability and selective permutation averaging. (a) Error rate under the original ordering for items receiving the same answer in all four cyclic rotations and for those that do not; Wilson intervals, with stability rates given under each label. (b) Accuracy gain over a single evaluation against mean inference calls per item, plotted as a gain so that two datasets 22 pp apart in absolute accuracy share one scale. Items are selected for additional permutations least-confident first, and compared against random selection at the same budget (averaged over 400 draws) and against an oracle that selects, under a fixed budget, the items where averaging helps.

Limitations. We evaluate one proprietary model, so these results do not generalise to System-1 models as a class; the model is closed-weight and exposes no seed, so exact reproduction depends on vendor behaviour. We evaluate no biosafety or hazard-classification task, and multiple-choice questions are only a proxy for the knowledge they are meant to probe; label quality in biology-adjacent multiple-choice sets can be poor [18] and we did not audit it, nor can contamination from public benchmarks be excluded. We did not measure latency and did not run an external larger model as a fallback, so we make no claim about either. Cyclic evaluation covers four rotations rather than all $4!$ permutations and only the four-option WMDP suites, making the reported instability a lower bound; we ran no designed repeated-identical-call test; answer labels are assigned in positional order, so a label prior and a position prior cannot be separated; and options appear twice in each request, which we did not ablate. Returned probabilities are quantised to two decimals, and several LAB-Bench high-confidence subsets contain fewer than 25 items, so we report intervals throughout.

6. Conclusion

We audited a commercial non-generative System-1 model on 6,020 items drawn from biosecurity-relevant and biology-relevant benchmarks. Its accuracy is strongly task-dependent, its top-1 probability is reasonably well calibrated and predicts its own errors well enough to support abstention, and its decisions are considerably less stable under reordering of the answer options than aggregate accuracy suggests. Averaging probabilities over answer permutations improves accuracy where instability is high, and applying it only to low-confidence items recovers most of that improvement at a fraction of the cost. Confidence and stability under

re-representation provide complementary evidence about when such a model’s predictions should be trusted, and neither is visible in a single accuracy figure.

Acknowledgements and Ethics

We thank TypeSafe for API access; the vendor had no role in the design, analysis, or conclusions of this study and did not review this manuscript before submission. This study authored no new hazardous technical content: all items come from existing public benchmarks, and we release model decision records but no new question content. We reported the confidence field semantics to the vendor as a documentation issue. All decision records and analysis code are released so that the reported statistics can be recomputed.

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LLM Usage Statement

An LLM-based coding assistant (Anthropic Claude) was used to write the evaluation harness and analysis scripts, to assist with statistical analysis and literature search, and for editorial work on this manuscript. All generated code was executed and its outputs inspected

by the authors; every statistic reported here is emitted by the released analysis pipeline rather than transcribed by hand, and the manuscript build fails if a number appears in the text that the pipeline did not produce. All bibliographic entries were verified against primary sources. An earlier version of this analysis contained statistical errors that were identified and corrected during an internal adversarial review; the corrected treatments are those reported in Section IV. The model under evaluation is itself a commercial closed-weight system, which is a reproducibility limitation we cannot remove. The authors take full responsibility for the correctness, originality, and integrity of all content.